B. TECH (CSE (Data Science)

sr.no.	Code	Course Title	L	T	P	Credit
1.	153601	Big Data & Tools	3	0	0	3
2.	153602	Data Visualization	3	0	0	3
3.	153603	Image and Video Processing	3	0	0	3
4.	153604	AI and Ethics	1	0	0	1
5.	153605	Constitution of India	3	0	0	0
6.	1536XX	Program Elective Course-II	3	0	0	3
7.	1536XX	Program Elective Course-III	3	0	0	3
8.	153601P	Big Data & Tools Lab	0	0	4	2
9.	153602P	Data Visualization Lab	0	0	4	2
10.	153603P	Image and Video Processing Lab	0	0	2	1
11.	153606P	NPTEL Course-II Lab	0	0	4	2
			TOTAL			23

List of Program Elective Courses-II

sr.no.	Code	Course Title	L	T	P	Credit
1.	153607	Compiler Design	3	0	0	3
2.	153608	Cyber Security	3	0	0	3
3.	153609	Deep Learning Algorithms	3	0	0	3
4.	153610	Network Security	3	0	0	3
5.	153611	Soft Computing	3	0	0	3

List of Program Elective Courses-III

sr.no.	Code	Course Title	L	T	P	Credit
1.	153612	Adversarial Machine Learning	3	0	0	3
2.	153613	Distributed System Design	3	0	0	3
3.	153614	Introduction to Wireless Networks	3	0	0	3
4.	153615	Intrusion Detection	3	0	0	3
5.	153616	Natural Language Processing	3	0	0	3
6.	153617	Reinforcement Learning	3	0	0	3

B. Tech (CSE (Data Science) **Semester- VI**

Course Code- 153601 Big Data and Tools 3 0 0 3

Unit-1.0: Introduction to Big Data 8 hrs

Big Data – Definition, Characteristic Features – Big Data Applications - Big Data vs Traditional Data - Risks of Big Data - Structure of Big Data - Challenges of Conventional Systems - Web Data – Evolution of Analytic Scalability - Evolution of Analytic Processes, Tools and methods - Analysis vs Reporting - Modern Data Analytic Tools.

Unit-2.0: Hadoop 6 hrs

Introduction: History of Hadoop - Hadoop Ecosystem - Analyzing data with Hadoop - Hadoop Distributed File System - Design - HDFS concepts - Hadoop filesystem - Data flow - Hadoop I / O - Data integrity - Serialization - Setting up a Hadoop cluster - Cluster specification - cluster setup and installation - YARN.

Unit-3.0: Map Reduce 6 hrs

Introduction: Understanding MapReduce functions - Scaling out - Anatomy of a MapReduce Job Run - Failures - Shuffle and sort - MapReduce types and formats - features - counters - sorting - MapReduce Applications –Configuring and setting the environment - Unit-1.0:test with MR Unit-1.0:- local test.

Unit-4.0: Spark 6 hrs

Installing spark - Spark applications - Jobs - Stages and Tasks - Resilient Distributed databases - Anatomy of a Spark Job Run - Spark on YARN - SCALA: Introduction - Classes and objects - Basic types and operators - built-in control structures - functions and closures - inheritance.

Unit-5.0: Big Data Frameworks 10 hrs

Matrix-Vector Multiplication – Hadoop YARN, Spark. Aggregate Data Models – Hbase: Data Model and Implementations – Hbase Clients – Examples – .Cassandra: Data Model – Examples – Cassandra Clients – Hadoop Integration. Pig – Grunt – Pig Data Model – Pig Latin – developing and testing Pig Latin scripts. Hive – Data Types and File Formats – HiveQL Data Definition – HiveQL Data Manipulation – HiveQL Queries.

Unit-6.0: NoSQL Databases 6 hrs

Introduction to NoSQL - MongoDB: Introduction - Data types - Creating - Updating and deleing documents - Querying - Introduction to indexing - Capped collections - Hbase: Concepts - Hbase Vs RDBMS - Creating records - Accessing data - Updating and deleting data - Modifying data - exporting and importing data. USE CASES: Call detail log analysis - Credit fraud alert - Weather forecast.

Text/Reference:

1. Bill Franks, ―Taming the Big Data Tidal Wave: Finding OpportUnities in Huge Data Streams with Advanced Analytics, Wiley and SAS Business Series, 2012.
2. David Loshin, "Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph", 2013.
3. Michael Berthold, David J. Hand, ―Intelligent Data Analysis‖, Springer, Second Edition, 2007.
4. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley, 2013.
5. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", Addison-Wesley Professional, 2012.
6. Richard Cotton, "Learning R – A Step-by-step Function Guide to Data Analysis, , O‘Reilly Media, 2013.

Course Code- 153602 Data Visualization 3 0 0 3

Unit-1.0: Introduction to Data and Types 7 hrs

Introduction to Data, The Basic Data Types – Nominal, Ordinal, interval, Ratio scaled. Non dependency Oriented Data: Quantitative Multidimensional Data, Categorical and Mixed Attribute Data, Binary and Set Data, Text Data.

Dependency-Oriented Data: Time-Series Data, Discrete Sequences and Strings, Spatial Data, Network and Graph Data.

Unit-2.0: Fundamentals of Data Visualization 7 hrs

The visualization pipeline, Visualizing Data: Mapping Data onto Aesthetics, Coordinate Systems and Axes, Colour scales, Directory of Visualizations, Visualizing Amounts.

Unit-3.0: Visualizing Distributions and Relationships 8 hrs

Visualizing Distributions and visualizing many Distributions at once, Visualizing Associations among Two or More Quantitative Variables, Visualizing Time series and other functions of an independent variable.

Unit-4.0: Advanced Visualization Techniques 7 hrs

Visualize Trends, Geospatial data and uncertainty, Visualization of Networks and Trees, visualizing multidimensional data, Data Reduction – Reduce Items and Attributes.

Unit-5.0: Principles and Challenges in Data Visualization 7 hrs

The principles of proportional link, Handling overlapping points, Balance the data and context, Understanding the most commonly used Image file formats, choosing the right visualization software.

Unit-6.0: Data Visualization and Graphical Analysis 6 hrs

Visualized exploratory data Analysis, Histograms and frequency polygons, Box-plots, Quartiles, Scatter Plots, Heat Maps.

Text/Reference:

1. Claus O. Wilke, Fundamentals of Data Visualization, O'Reilly publication, first Release Edition.
2. Chun-houh Chen, Wolfgang Hardle, Antony Unwin, Handbook of Data Visualization, Springer, 2008
3. Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014
4. Kieran Healy, Data Visualization: A Practical Introduction 1stEdition, Princeton university press.
5. Jiawei Han and Micheline Kamber, Data Mining- Concepts and Techniques-Morgan Kaufmann Publishers, Elsevier, 2nd Edition, 2006.

Course Code- 153603 Image and Video Processing 3 0 0 3

Unit-1.0: Fundamentals of Image Processing and Image Transforms 5 hrs

Digital Image fundamentals, Sampling and quantization of an Image, Relationship between pixels.
Image Transforms: 2-D Discrete Fourier Transform, Properties, Discrete cosine Transform, Hadamard Transform.

Unit-2.0: Image Processing Techniques 9 hrs

Image Enhancement: Spatial Domain methods: Histogram Processing, Fundamentals of Spatial filtering, smoothing spatial filters, Sharpening Spatial filters. Frequency Domain methods: Basics of filtering in frequency domain, image smoothing, image sharpening.
Color image processing: Introduction, Fundamentals of color image processing: Color models - RGB, CMY, YIQ, HSI, Pseudo color image processing - intensity slicing, gray level to color transformation, Basics of Full Color image processing.

Unit-3.0: Image Restoration 7 hrs

Degradation Model, Inverse Filtering, Least Mean Square Filters, Constrained Least Squares Restoration.

Unit-4.0: Image Compression 9 hrs

Image compression fundamentals – coding Redundancy, spatial and temporal redundancy. Compression models: Lossy and Lossless, Huffmann coding, Arithmetic coding, run length coding, transform coding, predictive coding , JPEG standards.

Unit-5.0: Basic Steps of Video Processing 6 hrs

Analog video, Digital video, Time varying image formation model, Geometric image formation, formation, sampling of video signal.

Unit-6.0: 2D Motion Estimation 6 hrs

Optical flow, Pixel based motion estimation; Region based Motion estimation, Multi resolution motion estimation, Application of motion estimation in video coding.

Text/Reference:

1. Gonzalez and Woods ,"Digital Image Processing ", 3rd edition , Pearson .
2. Yao wang, Joem Ostarmann and Ya – Qin Zhang, "Video processing and communication ",1st edition , PHI.
3. M. Tekalp ,"Digital video Processing", Prentice Hall International.

Course Code- 153604 AI and Ethics 3 0 0 3

Unit-1.0: 2 hrs

Introduction: Definition of morality and ethics in AI, AI, and society.

Unit-2.0: 2 hrs

Impact of AI: Impact on society, Impact on human psychology, Impact on the legal system, Impact on the environment and the planet, and Impact on trust.

Unit-3.0: 2 hrs

Ethical Initiatives in AI: International ethical initiatives, Ethical harms, and concerns, Case study: healthcare robots, Autonomous Vehicles, Warfare, and weaponization.

Unit-4.0: 4 hrs

AI Standards And Regulation: Model Process for Addressing Ethical Concerns During System Design, Transparency of Autonomous Systems, Data Privacy Process, Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems.

Unit-5.0:

Rob ethics: Social and Ethical Implication of Robotics: 2 hrs

Robotics- Ethics and Morality, Moral Theories-Ethics in Science and Technology, Ethical Issues in an ICT Society, Harmonization of Principles, Ethics and Professional Responsibility, Robotics Taxonomy.

Unit-6.0: 2 hrs

Challenges and Opportunities: Ethical issues in artificial intelligence, Societal Issues Concerning the Application of Artificial Intelligence in Medicine, decision-making role in industries, and National and International Strategies on AI.

Text/Reference:

1. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp and Alan Winfield," The ethics of artificial intelligence: Issues and initiatives", EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452– March 2020.
2. Patrick Lin, Keith Abney, George A Bekey," Robot Ethics: The Ethical and Social Implications of Robotics", The MIT Press- January 2014.
3. Towards a Code of Ethics for Artificial Intelligence (Artificial Intelligence: Foundations, Theory, and Algorithms) by Paula Boddington, November 2017.
4. Mark Coeckelbergh," AI Ethics", The MIT Press Essential Knowledge series, April 2020.

Course Code- 153605 **Constitution of India** **3 0 0 0**

Unit-1.0: **7 hrs**

Meaning of the constitution law and constitutionalism, Historical perspective of the Constitution of India, Salient features and characteristics of the Constitution of India.

Unit-2.0: **7 hrs**

Scheme of the fundamental rights, The scheme of the Fundamental Duties and its legal status, The Directive Principles of State Policy – Its importance and implementation.

Unit-3.0: 7 **hrs**

Federal structure and distribution of legislative and financial powers between the Union and the States, Parliamentary Form of Government in India – The constitution powers and status of the President of India.

Unit-4.0: **7 hrs**

Amendment of the Constitutional Powers and Procedure, The historical perspectives of the constitutional amendments in India.

Unit-5.0: **7 hrs**

Emergency Provisions: National Emergency, President Rule, Financial Emergency, Local Self Government – Constitutional Scheme in India.

Unit-6.0: **7 hrs**

Scheme of the Fundamental Right to Equality, Scheme of the Fundamental Right to certain Freedom under Article 19, Scope of the Right to Life and Personal Liberty under Article 21.

Text/Reference:

1. "Constitution of India" (for Competitive Exams) - Published by Naidhruva Edutech Learning Solutions, Bengaluru. – 2022.
2. "Introduction to the Constitution of India", (Students Edition.) by Durga Das Basu (DD Basu): Prentice –Hall, 2008.
3. "Constitution of India, Professional Ethics and Human Rights" by Shubham Singles, Charles E. Haries, and et al: published by Cengage Learning India, Latest Edition – 2019.
4. "The Constitution of India" by Merunandan K B: published by Merugu Publication, Second Edition, Bengaluru.
5. "Samvidhana Odu" - for Students & Youths by Justice HN Nagamohan Dhas, Sahayana, kerekon.
6. M.Govindarajan, S.Natarajan, V.S.Senthilkumar, "Engineering Ethics", Prentice –Hall, 2004.

Program Elective-II

Course Code- 153607 Compiler Design 3 0 0 3

Unit-1.0: 7 hrs

Introduction: Phases of compilation and overview.

Lexical Analysis (scanner): Regular languages, finite automata, regular expressions, from regular expressions to finite automata, scanner generator (lex, flex).

Unit-2.0: 9 hrs

Syntax Analysis (Parser): Context-free languages and grammars, push-down automata, LL(1) grammars and top-down parsing, operator grammars, LR(O), SLR(1), LR(1), LALR(1) grammars and bottom-up parsing, ambiguity and LR parsing, LALR(1) parser generator (yacc, bison).

Unit-3.0: 8 hrs

Semantic Analysis: Attribute grammars, syntax directed definition, evaluation and flow of attribute in a syntax tree.

Symbol Table: Its structure, symbol attributes and management. Run-time environment: Procedure activation, parameter passing, value return, memory allocation, and scope.

Unit-4.0: 7 hrs

Intermediate Code Generation: Translation of different language features, different types of intermediate forms.

Code Improvement (optimization) Analysis: control-flow, data-flow dependence etc.; Code improvement local optimization, global optimization, loop optimization and peep-hole optimization.

Unit-5.0: 4 hrs

Architecture dependent code improvement: instruction scheduling (for pipeline) and loop optimization (for cache memory). Register allocation and target code generation.

Unit-6.0: 7 hrs

Advanced topics: Type systems, data abstraction, compilation of Object Oriented features and non-imperative programming languages.

Text/Reference:

1. Compilers Principles Techniques And Tools by Alfred V. Aho, Ravi Sethi, Jeffery D. Ullman. Pearson Education.
2. Compiler Design by Santanu Chattopadhyay. PHI.
3. Modern Compiler Design by Dick Grune, E. Bal. Ceriel, J. H. Jacobs, and Koen G. Langendoen, Viley Dreamtech.

Course Code- 153608 Cyber Security 3 0 0 3

Unit-1.0: Cyber Security Fundamentals 7 hrs

Introduction, cybercrime and Information Security, Classification of Cybercrimes, Cybercrime the legal Perspective, Cyberoffences,, Cyber-attacks, Social engineering, Cyberstalking, Botnets, Attack Vector, Cybercrime and cloud computing, Cybercrimes on mobile and wireless devices, Attacks on mobile/cell phones.

Unit-2.0: Authentication and Security Measures 9 hrs

Authentication service security, Organizational security policies and measures in mobile computing.

Unit-3.0: Ethics in Cyber Security 8 hrs

Privacy, Intellectual Property in cyberspace, Professional Ethics, Freedom of Speech, Fair User and Ethical Hacking, Trademarks, Internet Fraud, Electronic Evidence, forensic Technologies, Digital Evidence collection.

Unit-4.0: Tools and Methods Used in Cybercrime 7 hrs

Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Phishing and Identity Theft, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer overflows.

Unit-5.0: Cybercrimes and Cyber security 4 hrs

Cybercrime and Legal Landscape around the world, Cyber laws, The Indian IT Act, Challenges, Digital Signatures and Indian IT Act, Amendments to the Indian IT Act, Cybercrime and punishment, Cost of Cybercrimes and IPR Issues, Web threats for Organizations, Social Computing and associated Challenges for Organizations.

Unit-6.0: Cybercrime Examples and Mini-Cases 7 hrs

Career Paths in Cyber security, Honeypots, Case study (Official Website Hacking, E-mail spoofing, Banking related Frauds, Credit Card related Frauds).

Text/Reference:

1. Cyber Security by Nina Godhole, SUnit-1.0:Belapure, Wiley India.
2. Cyber Security Essentials by James Graham, Ryan Olson, Rick Howard CRC Press, Taylor & Francis Group, 2011 ISBN: 978-1-4398-5123-4.

Course Code- 153609 Deep Learning Algorithms 3 0 0 3

Unit-1.0: 4 hrs

Introduction to Neural Network: Brief introduction to neural networks. Error Cost Function, Mean-Square Error, Cross-Entropy. Linear vs Nonlinear functions: Activation Function – Linear, Sigmoid, Softmax, Relu.

Unit-2.0: 4 hrs

Optimization: Gradient Descent Algorithm, Momentum Based Gradient Descent Algorithm and Nesterov Accelerated Gradient Descent Algorithm.

Unit-3.0: 9 hrs

Algorithm for Pattern Association/Reconstruction: Associative memory networks, Hetero Associative Memory neural networks, Auto Associative Memory Networks, Bi-directional Associative memory networks.

Feedback Networks: Discrete Hopfield Net, Continuous Hopfield Net, Radial Basis Function Networks (RBFN): Training algorithm for an RBFN with fixed Centres.

Unit-4.0: 9 hrs

Cluster Network: Self-Organizing Feature Map, Plasticity/Stability Problem, Adaptive Resonance Theory (ARTI and ARTII) - Architecture, ARTMAP.

Dense Embeddings: word2vec – Skip-gram model, Continuous Bag Of Words (CBOW) Model.

Unit-5.0: 8 hrs

Convolutional Neural Network: Architecture, Convolution Operation, Filters, Pooling Operation – Max pooling, Average pooling, Global pooling. Dilated CNN.

Unit-6.0: 8 hrs

Deep Recurrent Neural Network: Long Short Term Memory (LSTM) – Architecture. Gated Recurrent Unit-1.0:(GRU). Attention Mechanism. Recursive Neural Network.

Deep Unsupervised Learning: Autoencoders, Variational Autoencoders, Adversarial Generative Networks.

Text/Reference:

1. Neural Network by Simon Haykin, Pearson Education/PHI.
2. Deep Learning, Part II. Goodfellow, I., Bengio, Y., Courville, A., MIT Press, 2016.
3. Learning deep architectures for AI. Bengio, Yoshua. Foundations and trends in Machine Learning 2.1 (2009): 1127.

Course Code- 153610 Network Security 3 0 0 3

Unit-1.0: 4 hrs

Computer Security concepts, OSI security Architecture, Security attacks, Security Services, security Mechanisms, model of network security, standards, Physical Layer Security, network device security.

Unit-2.0: 6 hrs

Symmetric encryption Principles, Block cipher algorithms, random and pseudorandom numbers, stream ciphers and RC4, block cipher modes, secure hash functions, message authentication codes, public-key-cryptography principles, digital signature, access control.

Unit-3.0: 8 hrs

Key distribution and user authentication, Kerberos, X.509 Certificates, federated identity management, web security considerations, Secure Socket layer, transport layer security, HTTPS 160, Secure Shell(SSH), IEEE 802.11i Wireless LAN Security, wireless application protocol overview, Wireless transport layer security, WAP end-to-end security.

Unit-4.0: 8 hrs

Electronic mail security , Pretty good privacy, S/MIME, Domain keys Identified Mail, IP Security overview, IP security Policy, Encapsulating security payload, combining security association, Internet key exchange, DNSSec, eSMTPS, TCP session hijacking, ARP attacks, route table modification, UDP hijacking, and man-in-the-middle attacks.

Unit-5.0: 6 hrs

System Security- Types of malicious software, viruses, worms, DoS, DDoS, BOF, countermeasures, IDS concepts, IDS types and detection models, IDS features, IDS deployment considerations, Firewalls characteristics, Types of Firewalls, Firewall location and Configuration.

Unit-6.0: 10 hrs

Voice over IP (VoIP) and PBX security, VoIP components, VoIP vulnerabilities and countermeasures, securing a PBX, Proxy or application level gateways as security devices, VPN, Privacy protection and anonymity services, Electronic payment system.

Text/Reference:

1. “Network Security Essentials: Applications and Standards” by William Stallings, Pearson.
2. “Network Security private communication in a public world”, C. Kaufman, R. Perlman and M. Speciner, Pearson.
3. LAN Switch Security: What Hackers Know About Your Switches, 1st Ed. (2007), by Eric Vyncke and Christopher Paggen.
4. “Designing Network Security”, Merike Kaeo, 2nd Edition, Pearson Books.
5. “Building Internet Firewalls”, Elizabeth D. Zwicky, Simon Cooper, D. Brent Chapman, 2nd Edition, Oreilly publisher.
6. “Practical Unix & Internet Security”, Simson Garfinkel, Gene Spafford, Alan Schwartz, 3rd Edition, O’reilly publisher.

B. Tech (CSE (Data Science) **Semester- VI**

Course Code- 153611 **Soft Computing** **3 0 0 3**

Unit-1.0 **8 hrs**

Introduction to Soft Computing: Evolution of Computing: Soft Computing Constituents, Soft vs Hard Computing, From Conventional AI to Computational Intelligence: Machine Learning Basics.

Unit-2.0 **6 hrs**

Neural Networks: Machine Learning Using Neural Network, Adaptive Networks, Feed forward Networks, Supervised Learning Neural Networks,

Unit-3.0 **6 hrs**

Radial Basis Function Networks: Reinforcement Learning, Unsupervised Learning Neural Networks, Adaptive Resonance architectures, Advances in Neural networks.

Unit-4.0 **8 hrs**

Fuzzy Systems: Fuzzy Sets, Operations on Fuzzy Sets, Fuzzy Relations, Membership Functions: Fuzzy Rules and Fuzzy Reasoning, Fuzzy Inference Systems, Fuzzy Expert Systems, Fuzzy Decision Making, Fuzzy based Back Propagation network.

Unit-5.0 **7 hrs**

Genetic Algorithm: Introduction to Genetic Algorithms (GA), Applications of GA in Machine Learning: Machine Learning Approach to Knowledge Acquisition, GA based Back P.

Unit-6.0 **7 hrs**

Matlab/Python Lib: Introduction to Matlab/Python, Arrays and array operations, Functions and Files, Study of neural network toolbox and fuzzy logic toolbox, Simple implementation of Artificial Neural Network and Fuzzy Logic.

Text/ Reference:-

1. Principles Of Soft Computing, 2nd Ed (With CD) Book by S. N. Deepa and S. N. Sivanandam
2. Soft Computing, D. K. Pratihar, Narosa, 2008.
3. Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis & Applications, S.Rajasekaran, G. A. Vijayalakshami, PHI.
4. Genetic Algorithms: Search and Optimization, E. Goldberg.
5. Neuro-Fuzzy Systems, Chin Teng Lin, C. S. George Lee, PHI
6. Build Neural Network With MS Excel sample by Joe choong.

Program Elective-III

Course Code-153612 **Adversarial Machine Learning** **3 0 0 3**

Unit-1.0: **6 hrs**

Overview of adversarial machine learning, Statistical machine learning. A framework for secure learning, security analysis, framework, exploratory attacks, causative attacks, repeated learning games, Privacy-Preserving learning.

Unit-2.0: **8 hrs**

Attacking a hypersphere learner, causative attacks on hypersphere detectors, hypersphere attack description, optimal unconstrained attacks, imposing time constraints on the attack, attacks against retraining with data replacements, constrained attackers.

Unit-3.0: **7 hrs**

Availability attack Case Study: SpamBayes, Spam Filter, threat model, causative attacks, RONI Defense, Experiments with SpamBayes.

Unit-4.0: **7 hrs**

Integrity Attack Case Study: PCA Detector, traffic anomalies, corrupting the PCA subspace, Corruption-Resilient Detectors, empirical evaluation.

Unit-5.0: **8 hrs**

Privacy-Preserving Mechanism for SVM learning, Privacy Breach case studies, problem setting: privacy-preserving learning, Support Vector Machines: A brief primer, differential privacy.

Unit-6.0: **6 hrs**

Near optimal evasion of classifiers, convex classes. Adversarial machine learning challenges, open problems, defensive technologies.

Text/References:

1. Adversarial Machine Learning by Anthony D. Joseph, Blaine Nelson, Benjamin I. P. Rubinstein, J. D. Tygar
2. Adversarial Machine Learning by Yevgeniy Vorobeychik, Murat Kantarcioglu

Course Code- 153613 Distributed System Design 3 0 0 3

Unit-1.0: 7 hrs

Introduction: Motivation for distributed systems; Characteristics of distributed systems and middleware categories; Network service requirements for distributed systems; Architectural models of distributed systems.

Interprocess communication: Multithreading models and multithreaded client/server architectures; Hardware virtualization; Code mobility; Virtual machine migration; Interprocess communication paradigms in distributed systems.

Unit-2.0: 8 hrs

Naming: Naming scheme and namespace; URI, URN, URL; Name resolution and DNS naming service; Naming and location services of mobile entities; Distributed garbage collection.

Synchronization: Clock synchronization; NTP and GPS; Lamport's clock and vector clock; Total-ordered and causal-ordered multicasting; Election and mutual exclusion in distributed systems.

Coordination: Coordination models; Direct and indirect coordination; Gossip based and group coordination; Publish-subscribe coordination.

Unit-3.0: 7 hrs

Consistency: Consistency models; Data-centric and client-centric consistency; Linearizability; Sequential, causal and FIFO consistency; Eventual consistency; Monotonic read/write, read-your-writes, writes-follow-reads consistency.

Replication: Replication system architecture and functioning; Passive and active replication; Replication in gossip architecture; Consistency models and protocols in replication; Continuous consistency and conit; Replica management, placement and distribution.

Unit-4.0: 7 hrs

Fault Tolerance: Dependability; Failure model; Failure handling and resiliency; Reliable group communication; Distributed commit and consensus.

Failure Recovery: Forward-error and backward-error recovery; Rollback and checkpointing; Message logging.

Unit-5.0: 7 hrs

Distributed object-based and component-based systems: Distributed objects; Object servers and adapters; Distributed components and containers.

Distributed event-based systems: Distributed event-based system architectures; Content-based communication; Matching and filtering.

Unit-6.0: 6 hrs

Distributed web-based systems: Web service descriptions; SOAP and REST protocol; UDDI service; Web servers and web server clusters.

Distributed file systems: Distributed file system design requirements; File service architecture.

Pervasive systems: Sensor, actuator and IoT; Ubiquity; Sensor network; Context awareness.

Text/Reference:

1. Maarten van Steen and Andrew S. Tanenbaum, *Distributed Systems*, Third edition, 2017.
2. George Coulouris, Jean Dollimore, Tim Kindberg and Gordon Blair, *Distributed Systems: Concepts and Design*, Fifth edition, Pearson/Addison-Wesley, 2012.
3. Andrew S. Tanenbaum and Maarten van Steen, *Distributed Systems: Principles and Paradigms*, Second edition, Pearson/Prentice-Hall, 2007.

4. Joel M. Crichlow, *Distributed Systems: Computing Over Networks*, Second edition, Pearson/Prentice-Hall, 2014.
5. Sukumar Ghosh, *Distributed Systems: An Algorithmic Approach*, Chapman and Hall/CRC, 2014.
6. Arno Puder, Kay Römer and Frank Pilhofer, *Distributed Systems Architecture: A Middleware Approach*, Elsevier, 2011.
7. Ajay D. Kshemkalyani and Mukesh Singhal, *Distributed Computing: Principles, Algorithms, and Systems*, Cambridge University Press, 2008.
8. Pankaj Jalote, *Fault Tolerance in Distributed Systems*, Prentice-Hall, 1994.
9. Mukesh Singhal and Niranjan G. Shivaratri, *Advanced Concepts in Operating Systems*, McGraw-Hill, 1994.

Course Code- 153614 Introduction to Wireless Networks 3 0 0 3

Unit-1.0: Introduction 7 hrs

Overview of wireless and mobile infrastructure, Preliminary concepts on cellular architecture, Design objectives and performance issues, Radio resource management and interface, Propagation and path loss models, Channel interference and frequency reuse, Cell splitting, Channel assignment strategies, Overview of generations: - 1G to 5G.

Unit-2.0: Location Management 7 hrs

Introduction to location management (HLR and VLR), Mobility models characterizing individual node movement (Random walk, Fluid flow, Markovian, Activity based), Mobility models characterizing the movement of groups of nodes (Reference point based group mobility model, Community based group mobility model), Static (Always vs. Never update, Reporting Cells, Location Areas) and Dynamic location management schemes (Time, Movement, Distance, Profile Based), Terminal Paging (Simultaneous paging, Sequential paging); Location management and Mobile IP.

Unit-3.0: Handoff Management 4 hrs

Overview of handoff process, Factors affecting handoffs and performance evaluation metrics, Handoff strategies, Different types of handoffs (soft, hard, horizontal, vertical).

Unit-4.0: Basics of Wireless Transmission 9 hrs

Introduction to narrow and wideband systems, Spread spectrum; Frequency hopping, Introduction to MIMO, MIMO Channel Capacity and diversity gain, Introduction to OFDM, MIMO-OFDM system, Multiple access control (FDMA, TDMA, CDMA, SDMA), Wireless local area network, Wireless personal area network (Bluetooth and Zigbee).

Unit-5.0: Cognitive Radio Networks 7 hrs

Fixed and dynamic spectrum access, Direct and indirect spectrum sensing; Spectrum sharing, Interoperability and co-existence issues, Applications of cognitive radio networks.

Unit-6.0: Basics Of 5g Networks 7 hrs

Introduction to D2D communications; High level requirements for 5G architecture, Introduction to the radio resource management, power control and mode selection problems, Milli-meterwave communication in 5G.

Text/Reference Books

1. Jochen Schiller, Mobile Communications, 2nd Edition, Pearson Education, 2008 , Pearson Education
2. Andrea Goldsmith, Wireless Communications, 1st Edition, Cambridge University Press, 2009, 978-0521704168.

Course Code-153615 **Intrusion Detection** **3 0 0 3**

Unit-1.0: **8 hrs**

Network Attacks, Understanding Intrusion Detection and Intrusion Prevention System, Detection Approaches (Misuse Detection, Anomaly Detection, etc.), Uses of IDPS Technologies, Key Functions of IDPS Technologies, Stateful Protocol Analysis.

Unit-2.0: **9 hrs**

Data Collection (Host-Based, Network-Based, Application-Based, Application-Integrated and Hybrid), Theoretical Foundations of Detection - Taxonomy of anomaly detection system, fuzzy logic, Bayes theorem, Artificial Neural networks, Support vector machine, Evolutionary computation, Association rules, Clustering, Architecture and Implementation.

Unit-3.0: **9 hrs**

IDS Challenges, Alert Management & Correlation (Data Fusion, Alert Correlation, Cooperative Intrusion Detection), Evaluation Criteria- Accuracy, Performance, Completeness, Timely Response, Adaptation and Cost-Sensitivity, Intrusion Tolerance and Attack Resistance, Test. Intrusion Response.

Unit-4.0: **6 hrs**

Security and IDS Management (Data Correlation, Incident Response, Policy and Procedures, Law, Standards and organizations, Security Business issues, Future of Intrusion Detection and Prevention).

Unit-5.0: **6 hrs**

Implementation and Deployment: Internet Security System's Real Source, Snort, NFR Security, IDS Tools. Detailed case study of IDS in different networks like Ethernet Networks, 802.11 Networks, Mobile Networks, Ad-hoc Networks, and Wireless Sensor Networks.

Unit-6.0: **4 hrs**

Introduction to Snort, Different modes of Snort, Snort IDS Components, Snort Rules, Snort Filters, Snort output, Alert modes.

Text/ Reference:

1. Network Intrusion Detection and Prevention by Ali A. Ghorbani, Wei Lu Mahbod Tavallaee, Springer.
2. Intrusion Detection & Prevention by Carl Endorf, Eugene Schultz, and Jim Mellander, TMH.
3. Implementing Intrusion Detection Systems by Tim Crothers, Wiley.

Course Code- 153616 **Natural Language Processing** **3 0 0 3**

Unit-1.0: **7 hrs**

Sound: Biology of Speech Processing; Place and Manner of Articulation; Word Boundary Detection; Argmax based computations; HMM and Speech Recognition.

Unit-2.0: **7 hrs**

Words and Word Forms: Morphology fundamentals; Morphological Diversity of Indian Languages; Morphology Paradigms; Finite State Machine Based Morphology; Automatic Morphology Learning; Shallow Parsing; Named Entities; Maximum Entropy Models; Random Fields.

Unit-3.0: **7 hrs**

Structures: Theories of Parsing, Parsing Algorithms; Robust and Scalable Parsing on Noisy Text as in Web documents.

Unit-4.0: **7 hrs**

Hybrid of Rule Based and Probabilistic Parsing; Scope Ambiguity and Attachment Ambiguity resolution.

Unit-5.0: **7 hrs**

Meaning: Lexical Knowledge Networks, Wordnet Theory; Indian Language Wordnets and Multilingual Dictionaries; Semantic Roles; Word Sense Disambiguation; WSD and Multilinguality; Metaphors; Coreferences.

Unit-6.0: **7 hrs**

Web 2.0 Applications: Sentiment Analysis; Text Entailment; Robust and Scalable Machine Translation; Question Answering in Multilingual Setting; Cross Lingual Information Retrieval (CLIR).

Text/ Reference:

1. Jurafsky, Dan and Martin, James, “Speech and Language Processing”, 2nd Edition, Prentice Hall, 2008
2. Manning, Christopher and Heinrich, Schutze, “Foundations of Statistical Natural Language Processing”, MIT Press, 1999.
3. Allen James, “Natural Language Understanding”, 2nd edition, Benjamin Cumming, 1995.
4. Charniack, Eugene, “Statistical Language Learning”, MIT Press, 1993.

Course Code- 153617 Reinforcement Learning 3 0 0 3

Unit-1.0: Introduction to Reinforcement Learning Problem 4 hrs
Reinforcement Learning, Elements of Reinforcement Learning, Limitations and Scope, An Extended Example: Tic-Tac-Toe, History of Reinforcement Learning.

Unit-2.0: Multi-arm Bandits 5 hrs
An n-Armed Bandit Problem, Action-Value Methods, Incremental Implementation, Tracking a Nonstationary Problem, Optimistic Initial Values, Upper-Confidence-Bound

Unit-3.0: Finite Markov Decision Processes 8 hrs
The Agent–Environment Interface, Goals and Rewards, Returns, Unified Notation for Episodic and Continuing Tasks, The Markov Property, Markov Decision Processes, Value Functions, Optimal Value Functions, Optimality and Approximation.

Unit-4.0: Dynamic Programming 7 hrs
Policy Evaluation, Policy Improvement, Policy Iteration, Value Iteration, Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming.

Unit-5.0: BMonte Carlo Methods 9 hrs
Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control without Exploring Starts, Off-policy Prediction via Importance Sampling, Incremental Implementation, Off-Policy Monte Carlo Control, Importance Sampling on Truncated Returns.

Unit-6.0: Temporal-Difference Learning and Policy Approximation 9 hrs
TD Prediction, Advantages of TD Prediction Methods, Optimality of TD(0), Sarsa: On-Policy TD Control, Q-Learning: Off-Policy TD Control, Games.
Actor–Critic Methods, Eligibility Traces for Actor–Critic Methods, R-Learning and the Average-Reward Setting, Vanilla policy gradient method.

Text/ Reference:

1. RS Sutton Reinforcement Learning: An Introduction – Stanford University.
2. Enes Bilgin, Mastering Reinforcement Learning with Python, Packt.
3. Sudharsan Ravichandiran, Hands-On Reinforcement Learning with Python: Master Reinforcement and Deep Reinforcement Learning Using OpenAI Gym and TensorFlow.
4. Aske Plaat, Deep Reinforcement Learning, Springer Singapore

B. Tech (CSE (Data Science) **Semester- VI**

Course Code- 153601P **Big Data and Tools Lab** **0 0 4 2**

Perform all experiments

List of experiments (at least 10, equally distributed across all topics):

1. **Exploring Big Data Characteristics (Python/Pandas):** Load a large dataset and analyze its characteristics based on the 5Vs (Volume, Velocity, Variety, Veracity, and Value).
2. **Comparing SQL and Big Data Processing (SQLite/PySpark):** Perform the same queries on a dataset using both SQLite and PySpark, and compare execution times.
3. **Modern Data Analytic Tools (Hadoop/Spark/Kafka):** Install and explore Hadoop, Spark, and Kafka. Write a basic PySpark program to process a text file.
4. **HDFS File Operations (Hadoop/MinIO):** Perform basic file operations (upload, download, list, delete) in HDFS or MinIO.
5. **Map Reduce Word Count Program (Java/Python MapReduce):** Implement a MapReduce program to count word occurrences in a text file.
6. **Matrix-Vector Multiplication Using MapReduce (Hadoop/PySpark):** Implement matrix-vector multiplication using MapReduce and execute it on Hadoop or PySpark.
7. **Hadoop YARN & Spark Execution (YARN/PySpark):** Submit a PySpark job on Hadoop YARN and monitor its execution.
8. **Predictive Analytics using Spark MLlib (PySpark MLlib):** Train and test a logistic regression model using Spark MLlib.
9. **NoSQL Database Operations with Cassandra (Cassandra/MongoDB):** Create a NoSQL database, insert data, and perform queries using Cassandra or MongoDB.
10. **Big Data Querying using Hive & Pig (HiveQL/Pig Latin):** Execute data manipulation and querying tasks using HiveQL and Pig Latin.

Course Code- 153602P **Data Visualization Lab** **0 0 4 2**

Perform any 10 experiments

List of experiments (at least 10, equally distributed across all topics):

1. **Exploring Different Data Types (Python/Pandas):** Load a dataset and classify its attributes into nominal, ordinal, interval, and ratio scales.
2. **Handling Non-Dependency Oriented Data (Python/Numpy):** Process and visualize different types of non-dependency data (e.g., categorical, binary, text) using NumPy and Pandas.
3. **Analyzing Dependency-Oriented Data (Python/Matplotlib):** Load and visualize time-series, spatial, or network data using appropriate plotting techniques.
4. **Understanding the Visualization Pipeline (Python/Seaborn):** Demonstrate the process of transforming raw data into visual insights using a structured visualization pipeline.
5. **Mapping Data onto Aesthetics (Python/Matplotlib/Seaborn):** Apply aesthetics such as color, shape, and size to visualize data attributes effectively.
6. **Using Different Coordinate Systems (Python/Matplotlib):** Compare Cartesian, polar, and logarithmic coordinate systems with different types of plots.
7. **Applying Color Scales in Visualizations (Python/Seaborn):** Create visualizations using various color palettes and explain their impact on interpretation.
8. **Visualizing Data Distributions (Python/Seaborn):** Create histograms, density plots, and box plots to analyze data distributions.
9. **Comparing Multiple Distributions (Python/Seaborn/Matplotlib):** Use violin plots and swarm plots to visualize multiple distributions simultaneously.
10. **Visualizing Relationships Between Variables (Python/Seaborn):** Create scatter plots, pair plots, and correlation matrices to analyze associations between variables.
11. **Visualizing Time-Series Data (Python/Matplotlib/Pandas):** Plot time-series data with trendlines and moving averages to observe patterns over time.
12. **Visualizing Geospatial Data (Python/Folium/GeoPandas):** Plot geospatial data on maps using Python libraries like Folium or GeoPandas.
13. **Network and Tree Visualization (Python/NetworkX/Matplotlib):** Generate and visualize a network graph or hierarchical tree structure.
14. **Handling Multidimensional Data (Python/Seaborn/PCA):** Reduce dimensionality using PCA and visualize high-dimensional data using pair plots or 3D scatter plots.
15. **Handling Overlapping Data Points (Python/Seaborn/Matplotlib):** Implement jittering, transparency, or faceting to reduce clutter in visualizations.
16. **Comparing Image File Formats for Visualization (Python/OpenCV):** Save a visualization in different formats (JPEG, PNG, SVG) and compare quality vs. size.
17. **Choosing the Right Visualization Tool (Python/Tableau/Power BI):** Compare different visualization tools like Matplotlib, Tableau, and Power BI for specific use cases.

Course Code- 153603P **Image and Video Processing Lab** **0 0 2 1**

Perform any 10 experiments

List of experiments (at least 10, equally distributed across all topics):

1. **Image Sampling and Quantization (Python/OpenCV/MATLAB/Octave):** Reduce image resolution (sampling) and change bit depth (quantization) to observe effects on image quality.
2. **Pixel Relationships and Neighborhood Operations (Python/OpenCV/MATLAB/Octave):** Implement 4-connected and 8-connected neighborhood pixel analysis, and compute pixel intensity differences.
3. **Histogram Equalization (Python/OpenCV/MATLAB/Octave):** Compute and visualize image histograms and apply histogram equalization for contrast enhancement.
4. **Spatial Filtering – Smoothing (Python/OpenCV/MATLAB/Octave):** Apply mean, Gaussian, and median filters to remove noise and compare results.
5. **Spatial Filtering – Sharpening (Python/OpenCV/MATLAB/Octave):** Implement Laplacian and high-pass filters to enhance edges in an image.
6. **Frequency Domain Filtering (Python/OpenCV/MATLAB/Octave):** Perform Fourier Transform, apply low-pass and high-pass filtering, and reconstruct the image in the spatial domain.
7. **Color Model Conversion (Python/OpenCV/MATLAB/Octave):** Convert an image between RGB, CMY, YIQ, and HSI color spaces and analyze the differences.
8. **Pseudo Color Processing (Python/OpenCV/MATLAB/Octave):** Implement intensity slicing and grayscale-to-color transformation techniques.
9. **Image Degradation and Restoration (Python/OpenCV/MATLAB/Octave):** Simulate image degradation (motion blur, Gaussian noise) and restore it using inverse filtering.
10. **Least Mean Square (Wiener) Filtering (Python/OpenCV/MATLAB/Octave):** Apply Wiener filtering to restore a noisy and blurred image.
11. **Lossless Compression – Huffman Encoding (Python/MATLAB/Octave):** Implement Huffman encoding and decoding for image compression.
12. **Lossy Compression – JPEG Analysis (Python/OpenCV/MATLAB/Octave):** Apply JPEG compression and analyze its effect on image quality by varying quality factors.
13. **Video Frame Extraction (Python/OpenCV/MATLAB/Octave):** Extract frames from a video file and display them sequentially.
14. **Motion Detection in Video (Python/OpenCV/MATLAB/Octave):** Implement background subtraction techniques to detect moving objects in a video.
15. **Optical Flow Estimation (Python/OpenCV/MATLAB/Octave):** Implement optical flow algorithms (e.g., Lucas-Kanade) to estimate motion in a video sequence.